

RAJARATA UNIVERSITY OF SRI LANKA

FACULTY OF TECHNOLOGY

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

ICT 3219 – Mobile Application Development

Practical Sheet – 02

Understand the purpose and use cases of different Android layouts

Task 01 – Relative Layout

Relative Layout positions each child relative to the parent or to other sibling views — using rules like "below this," "to the right of that," "aligned with this view's top," etc.

Create a New Layout XML File

1. Right-click on res > layout.
2. Click **New > Layout Resource File**.
3. Name the file, e.g., relative_layout.xml.
4. Choose Root Element: Relative Layout, etc. based on layout type.
5. Click **OK**.

relative_layout.xml

1. Go to res > layout > relative_layout.xml.
2. Switch to **Code view** (not Design view).
3. **Replace the existing code** with the provided Relative Layout code.

```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="24dp"
    android:background="#E3F2FD">

    <!-- Dashboard Title -->
    <TextView
        android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="My Dashboard"
        android:textSize="28sp"
        android:textStyle="bold"
        android:textColor="#0D47A1" />

    <!-- Settings Button aligned to end of title -->
    <Button
        android:id="@+id/settings"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Settings"
```

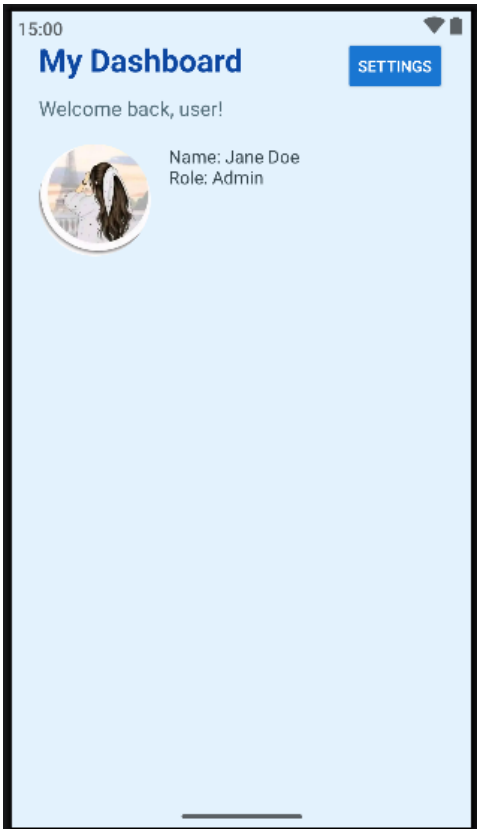
```
        android:textColor="#FFFFFF"
        android:backgroundTint="#1976D2"
        android:layout_alignBaseline="@id/title"
        android:layout_alignParentEnd="true" />

    <!-- Subtitle under the title -->
    <TextView
        android:id="@+id/subtitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome back, user!"
        android:textSize="18sp"
        android:textColor="#546E7A"
        android:layout_below="@id/title"
        android:layout_marginTop="12dp" />

    <!-- Profile Image below subtitle -->
    <ImageView
        android:id="@+id/profile_image"
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:layout_below="@id/subtitle"
        android:layout_marginTop="20dp"
        android:src="@drawable/profilepicture"
        android:layout_alignParentStart="true"
        android:background="@drawable/circle_bg"
        android:scaleType="centerCrop"
        android:clipToOutline="true" />

    <!-- User Info to the right of image -->
    <TextView
        android:id="@+id/user_info"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Name: Jane Doe\nRole: Admin"
        android:textSize="16sp"
        android:textColor="#37474F"
        android:layout_toEndOf="@id/profile_image"
        android:layout_alignTop="@id/profile_image"
        android:layout_marginStart="16dp" />

</RelativeLayout>
```



Task 02 – Table Layout

Table Layout arranges children into **rows and columns**, like an HTML table or spreadsheet. It's made up of Table Row elements — each Table Row represents one row, and each child inside that row becomes one column/cell.

Create a New Layout XML File

6. Right-click on res > layout.
7. Click **New > Layout Resource File**.
8. Name the file, e.g., table_layout.xml.
9. Choose Root Element: Table Layout, etc. based on layout type.
10. Click **OK**.

table_layout.xml

4. Go to res > layout > table_layout.xml.
5. Switch to **Code view** (not Design view).
6. **Replace the existing code** with the provided Table Layout code.

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#FAFAFA"
    android:padding="16dp"
    android:orientation="vertical">

    <!-- Title -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="📅 My Weekly Timetable"
        android:textSize="24sp"
        android:textStyle="bold"
        android:textColor="#3E7D80"
        android:layout_gravity="center_horizontal"
        android:layout_marginBottom="16dp" />
```

```
<!-- Timetable Table -->
<TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:stretchColumns="1"
    android:background="#00000000"
    android:elevation="4dp"
    android:padding="12dp">

    <!-- Header Row -->
    <TableRow android:background="#3E7D80">
        <TextView
            android:text="Time"
            android:textStyle="bold"
            android:textSize="16sp"
            android:textColor="#FFFFFF"
            android:padding="10dp" />

        <TextView
            android:text="Subject"
            android:textStyle="bold"
            android:textSize="16sp"
            android:textColor="#FFFFFF"
            android:padding="10dp" />
    </TableRow>

    <!-- Row 1 -->
    <TableRow android:background="#F1F9E9">
        <TextView
            android:text="08:00 - 09:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />

        <TextView
            android:text="Mathematics"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
    </TableRow>

    <!-- Row 2 -->
    <TableRow android:background="#E3E2F0">
        <TextView
            android:text="09:00 - 10:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />

        <TextView
            android:text="Physics"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
    </TableRow>

    <!-- Row 3 -->
    <TableRow android:background="#FFFFE0">
        <TextView
            android:text="10:00 - 11:00"
            android:textSize="15sp"
            android:padding="10dp"
            android:textColor="#000000" />
```

```
            <TextView
                android:text="Chemistry"
                android:textSize="15sp"
                android:padding="10dp"
                android:textColor="#000000" />
        </TableRow>

    </TableLayout>

</LinearLayout>
```

